6 Solve the simultaneous equations

$$5x + y = 21$$
$$x - 3y = 9$$

(1) $5x + y = 21$ (2) $x - 3y = 9$

(1) × 3: $15x + 3y = 63$

(2): $x - 3y = 9$

Add: $16x = 72$

$x = \frac{72}{16} = 4.5$

1 mark
Opposite coefficients: add to eliminate y

Substitute into (1): $22.5 + y = 21$

$y = 21 - 22.5 = -1.5$

1 mark
Substitute the found x

Check (2): $4.5 - 3(-1.5) = 9$

1 mark
Both values correct

$x =$ 4.5

$y =$ −1.5

(Total for Question 6 is 3 marks)